MILITARY SPECIFICATION

SIGNAL, SMOKE AND ILLUMINATION, MARINE MK 13 MOD 0

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacture, assembly, and preparation for delivery of the Signal, Smoke and Illumination, Marine Mk 13 Mod 0 and the methods of examination and tests upon which product acceptance will be based. The signal consists of a metal cylinder closed at each end with a tear friction type igniter. One end contains a pyrotechnic candle (red) for night use. The other end contains a smoke mixture (orange) for day use. Plastic caps cover both ends. The signal is classified as Class C Explosive (Code of Federal Regulations 49 CFR 171-178) and Class II-C Explosive (Coast Guard Rules and Regulations for Military Explosives and Hazardous Munitions GG 108).

1.2 Definitions. The meaning of terms used in this specification are in 6.4.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-45662	Calibration System Requirements

FSC 1370

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

Naval Ordnance Systems Command (Code Ident 10001)

LD 165580	Signal, Smoke and Illumination, Marine Mk 13 Mod 0 (Assembly)
LD 255173	Signal Container Mk 3 Mod 0
LD 299377	Box, Shipping and Storage Signal (Assembly)

PUBLICATIONS

Air Force

TPO 1370-115-3432	Transportation Packaging Order

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Hazardous Materials Regulations, Department of Transportation

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(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite the latest edition and supplement.)

COAST GUARD RULES AND REGULATIONS

GG 108	Rules and Regulations for Military Explosives and Hazardous Munitions

MANUALS

Department of Defense

4145.26M	DOD Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

FORMS

NAVORD Form 1204(10-44)	Pyrotechnic Ammunition Manufacturing Data Card

(Application for copies should be addressed to the Naval Publications and Forms Center, 5891 Tabor Avenue, Philadelphia, Pa. 19120.)

3. REQUIREMENTS

3.1 Conformance to documents. The signal shall be manufactured, examined, tested, and packed in accordance with contract documents, LD 165580, and all documents listed thereon, and this specification.

3.2 Preproduction sample. An acceptable preproduction sample of signals shall be furnished by the contractor. The signals, their components, and all materials shall be of single type, grade, class, size, and composition and shall comply with all applicable provisions of this document. The sample shall be manufactured using the proposed full production methods. The preproduction sample shall not be applied as part of the quantity specified for delivery by the contract or order.

3.3 Performance and product characteristics. The signal shall meet the acceptance criteria for performance and product characteristics in the sequence indicated in table I.

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3.3.1 Sealing. The signal, when tested in accordance with 4.3.1.1, shall withstand a vacuum of 6.0 inches ±1.0 of mercury (24 inches of mercury absolute pressure) for a minimum period of 1 minute without leakage.

3.3.2 Function. The signal, when tested in accordance with 4.3.1.2, shall meet the following requirements:

(a) Produce orange smoke and red flare from the designated end, as applicable.

(b) Pull strip shall separate from cover (both ends).

(c) Pull ring shall not separate from pull strip (both ends).

(d) Smoke burning limits - 10 seconds minimum to 26 seconds maximum.

(e) Flare burning limits - 12 seconds minimum to 28 seconds maximum.

3.4 Safety precautions. The supplier shall observe the safety precautions and requirements of DOD 4145.26M.

3.5 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaiminated by the cleaning agents employed.

4. QUALITY ASSURANCE PROVISIONS .

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and gages. All test equipment and gages shall be supplied and maintained by the contractor as specified in MIL-I-45607.

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The contractor shall maintain a calibration system as specified in MIL-C-45662.

4.1.2 Test and examinations classification. The tests and examinations herein are classified in regard to relative importance as critical, major, or minor. Tests and examinations classified as critical are identified by the symbol (C) and major by (M). The number following the symbol indicates the serial number of the test or examination. Tests which are not annotated are classified minor.

4.2 Sampling.

4.2.1 Preproduction sample. The contractor shall deliver a sample of 200 signals to the destination specified (see 6.2) for examination, testing, evaluation, and acceptance as specified in table I. Any production of the item prior to authorization by the procuring agency shall be at the contractor's risk. The sample shall be prepared for delivery as specified in section 5, except sufficient dunnage or spacers may be added to assure a firm pack, as applicable.

Table I

INSPECTION PLANS

Examinations and tests	Preproduction sampling plans		Production lot sampling plans	
1 2 3 Test sequence	Sample size	Acceptance criteria[1]	Sample size	Acceptance criteria[1]
Examination and workmanship (3.5) (4.4)	200 signals	Critical Ac 0 Re 1 Major Ac 5 Re 6 Minor Ac 10 Re 11	See para 4.4	See para 4.4
Sealing test (4.3.1.1) Function test[2] (4.3.1.2)	200 signals	Ac 0 Re 1 (a, b, & c) Ac 5 Re 6	MIL-STD-105 inspection level I	Ac 0 Re 1 (a, b, & c) AQL 1.0%
	MIL-STD-414 Code ltr Q (200 signals)	(d) AQL 1.0% (e) AQL 1.0%	MIL-STD-414 inspection level IV	(d) AQL 1.0% (e) AQL 1.0%

[1]Ac = acceptance number for specified requirements of section 3; Re = rejection number for specified requirements of section 3.
[2]Use the recorded data from units in the consecutive order of testing for requirements (d) and (e).

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4.2.2 Production lot. A production lot shall consist of 501 to 10,000 signals offered for delivery at one time including test samples. (See 6.2.) Sampling, examination, testing, and acceptance of production lots shall be in accordance with table I.

4.3 Tests. Random samples of signals shall be selected and tested as specified in 4.2, as applicable.

4.3.1 Performance tests.

4.3.1.1 Sealing test (M101). The signal shall be tested in a water tank equipped for observation under vacuum. The water temperature shall be 45° Fahrenheit (F) minimum to 75° F maximum.

CAUTION

The water tank should be equipped so that the signal must be raised above the water level while still under vacuum before releasing the vacuum and removing the signal.

The uppermost surface of the signal, when in a horizontal position, shall be submerged a minimum of 5 inches below the water surface and subjected to a vacuum of 6 inches ±1 of mercury (24 inches of mercury absolute pressure) for a minimum period of 1 minute. Leakers are indicated by air bubbles issuing from the signal. Signals which are not defective may be used for further testing or returned to the lot. Defectives are signals failing to meet the requirements of 3.3.1.

4.3.1.2 Function test (M102). The signals, prior to testing, shall be immersed under a 1-inch head of water at 68° F ± 5° F for a minimum period of 24 hours. The signals shall be function tested in accordance with the procedure specified on the instruction decal, Drawing 2113132. Defectives are signals failing to meet the acceptance criteria of table I for requirements specified in 3.3.2.

4.4 Examinations.

4.4.1 Signal examination (M103). Examinations shall be made to ascertain that the signals meet the requirements listed on Drawing 712793. Only signals containing no defective classified characteristics shall be used for further testing or shall be returned to the lot.

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4.4.2 Packaging, packing, and marking examination (C1). The signal shall be examined for conformance to the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. In addition to the following, level A packaging shall be in accordance with 49 CFR 171-178 for Class C Explosives.

5.1.1.1 Twelve signals shall be packaged in accordance with LD 255173. For Air Force shipments, 18 signals shall be packaged in a M2A1 metal box in accordance with Transportation Packaging Order (TPO) 1370-115-3432.

5.2 Packing.

5.2.1 Level A. In addition to the following, level A packing shall be in accordance with 49 CFR 171-178 for Class C Explosive.

5.2.1.1 One hundred eight signals packaged as in 5.1 shall be packed in a wood box in accordance with LD 299377. For Air Force shipments, 108 signals packaged in M2A1 metal boxes as described in 5.1 shall be packed in a wirebound box in accordance with Transportation Packaging Order (TPO) 1370-115-3432.

5.3.1 Special marking. In addition to the requirements of the contract or order, the outer container shall be marked in accordance with 49 CFR 171-178 for Class C Explosives (Hand Signal Devices - Handle Carefully - Keep Fire Away).

5.3.2 Normal marking. In addition to any requirements of the contract or order, packages and packs shall be marked in accordance with MIL-STD-129.

5.4 Data cards. The contractor shall fill out NAVORD Form 1204(10-44) 8-inch data cards made for each lot, if applicable (see 6.2). The contractor shall fill in all required information, including information for the following components:

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Component

Pull Wire Assembly
Smoke Candle Assembly
Flare Candle Assembly

6. NOTES

6.1 Intended use. The signal is used by aircraft personnel down at sea for day (orange smoke) or night (red flare) signaling.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, symbol, and date of this specification

(b) Quantity of signals being ordered

(c) Production lot size, if different from 4.2.2

(d) Test facility where preproduction samples are to be shipped for evaluation

(e) QE Laboratory coordinating surveillance

(f) Data information required (see 5.4).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraph below, (b) applicable documents listed in section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless justified and specified in the contract or order. The data to be furnished shall be as listed on DD Form 1423, which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 shall be attached where applicable.

6.3.1 The Government inspector shall endorse the cards and distribute one each as follows:

(a) Procuring activity

(b) SPCC (Code 730, Mechanicsburg, Pa., if different than the procuring activity)

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(c) QE Laboratory coordinating surveillance

(d) The designated test activity (see 6.2).

6.4 Definitions.

6.4.1 Signal. The term "signal" means Signal, Smoke and Illumination, Marine Mk 13 Mod 0 as depicted on Drawing 712793.

6.5 Conflicting requirements. Refer in writing to the procuring agency for clarification, any conflicting requirements arising between this specification and any other specification, publication, or drawing listed herein.

Custodians:
Navy - OS
Air Force - 70

Review activities:
Navy - MC
Air Force - 70

Preparing activity:
Navy - OS
(Project no. 1370-N311)

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